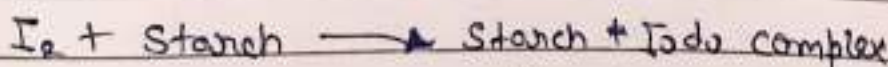
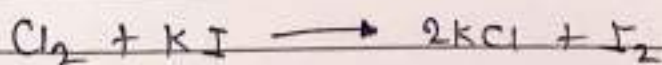


Object 1:- To determine the residual chlorine in a given water sample.

Apparatus and reagent:- Burette, pipette, Iodometric flask, Beakers, Dropper, Sodium thiosulphate ($\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$), potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$), potassium iodide (KI), Sulphuric Acid (Conc. H_2SO_4), Acetic acid (CH_3COOH), ~~di~~ starch.

Theory:- Sterilized water is produced for drinking purposes by passing chlorine gas through the sample for complete removal of micro-organisms. The germicide action of chlorine gas produces hypochlorous acid (HOCl) and nascent oxygen $[\text{O}]$. If chlorine in water is beyond certain limit, the water will become unfit for drinking. It is therefore, necessary to determine the amount of free chlorine in water sample.

The analysis is based on the oxidation of potassium iodide (KI) by the free chlorine present in water. The liberated iodine is estimated iodometrically by titrating it against standard hypo solution using starch as indicator.



Procedure :-(A) Standardization of Hypo solution -

- (i) Wash all the apparatus with distilled water.
- (ii) Fill the burette with Hypo solution.
- (iii) Pipette out 10 ml of $K_2Cr_2O_7$ solution becomes faint yellow.
- (iv) Add to it 10 ml H_2SO_4 to maintain pH and 5 ml KI solution.
- (v) Titrate it against hypo solution till the solution becomes faint yellow.
- (vi) Now add few drops of freshly prepared starch indicator, a blue colour appears. Continue to add hypo solution drop by drop while constantly shaking till the blue color disappears, it indicates the end point.

(B) Titration of water sample with Hypo solution -

- (i) Fill the burette with hypo solution.
- (ii) Take 10 ml of given water sample in a iodometric flask & add 3 ml glacial acetic.
- (iii) Add acid to attain pH level 3-4. Shake the solution & add 5 ml of KI solution.
- (iv) Titrate the water sample with standard hypo solution till the solution becomes faint yellow.
- (v) Now add few drops of freshly prepared starch indicator, a blue colour appears. Continue to add hypo solution drop by drop while constantly shaking till the blue color disappears, it indicates the end point.

Calculation:-

Titration between sample water with hypo solution

$$N_1 V_1 = N_2 V_2$$

Here N_1 = Normality of sample water

$$N_1 \times 50 = N_2 \times 30$$

N_2 = Normality of known Hypo

V_1 = Volume of sample water

$$N_1 = \frac{N \times V}{50 \times 30}$$

V_2 = Volume of known Hypo.

Strength of free and residual chlorine.

Here N_3 = Normality of $K_2Cr_2O_7$

$$N_3 V_3 = N_4 V_4$$

V_3 = Volume of $K_2Cr_2O_7$

$$N_3 = \frac{N_4 V_4}{V_3}$$

N_4 = Normality of Hypo solution

V_4 = Volume of ~~known~~ Hypo solution

$$N_3 = \frac{N \times V \times 30}{50 \times 30 \times 10}$$

$$N_3 = 0.0024 \text{ ml}$$

$$\begin{aligned} \text{Strength} &= N_3 \times \text{eq. weight} \times 1000 \\ &= 0.0024 \times 35.5 \times 1000 \\ &= 0.852 \times 1000 \\ &= 85.2 \text{ ppm} \quad \text{Answer} \end{aligned}$$

(vi) Note down the burette reading

(vii) Repeat the titration till the concordant reading is obtained.

Observation

Titration between $K_2Cr_2O_7$ & hypo solution

S. No	Volume of $K_2Cr_2O_7$ (ml)	Volume of hypo solution Initial (ml)	Final (ml)	Difference (ml)	Concordant reading (ml)
1.	10 ml	0.0	9.0	9.0	9.0
2.	10 ml	9.0	18.0	9.0	

Titration between K_2 water and hypo solution.

S.No	Volume of water Sample (ml)	Volume of hypo solution Initial (ml)	Final (ml)	Difference (ml)	Concordant reading (ml)
1.	50 ml	0	5	5	4.0
2.	50 ml	5	2.2	2.2	
3.	50 ml	7.2	11.2	4.0	

Strength of free and residual chlorine is ~~0.852~~

$$\begin{aligned} \text{Normality} \times \text{eq. weight of chlorine} \times 1000 \\ = 85.2 \text{ ppm} \end{aligned}$$

Result:- Amount of free & residual chlorine in the given water sample = 85.2 ppm.

Precautions:-

1. Freshly prepared starch solution should be used.
2. Distilled water should be used throughout the titration.
3. Starch solution be added just near end point.

Industrial Application:-

1. Chlorine determination is used to control chlorination of domestic and industrial waste water.
2. Chlorine determination is important to avoid bad odor & change in taste of water.
3. It is determined in the swimming pools to avoid ill effects due to excess chlorination.

Viva Question

Q1 What are iodometric titrations?

Ans Iodometric titration is a type of redox titration that involves iodine (I_2) & is used to determine the concentration of oxidizing agents. It is an indirect titration method where an oxidizing agent reacts with excess iodine ions to produce iodine (I_2) which is then titrated with a reducing agent like sodium thiosulphate ($Na_2S_2O_3$).

Q2 Differentiate between iodometric & iodimetric titration?

Ans Iodometric titration is an indirect method, where iodine is liberated by an oxidizing agent & then titrated.

Iodimetric titration is a direct method, where iodine itself is used as a titrant to react with a reducing agent.

Q3 Differentiate between primary & secondary solution.

Ans Primary solution are prepared from highly pure, stable compounds & have known concentration.

Secondary solution have unknown or opportunities concentrations and must be standardized before use in titration.

Q4 Which type of indicator is used in this titration? Name it?

Ans Starch.

Q5 What is the equivalent weight of copper sulphate?

Ans Equivalent weight of $\text{CuSO}_4 = 79.75 \text{ gm/e}$
 $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (Hydrated) 249.80 gm/e

Q6 Why freshly prepared starch solution is used in this titration?

Ans Because starch decomposes over time, leading to inaccurate result.

Teacher's Signature _____

Q7 Why starch is added near the end point of the titration?

Ans Starch is added near the end point of titration, because it acts as an indicator for the presence of iodine. If we add starch in beginning then starch-iodine complex becomes too stable & may not dissociate.

Q8 Explain the reaction involved in this titration?

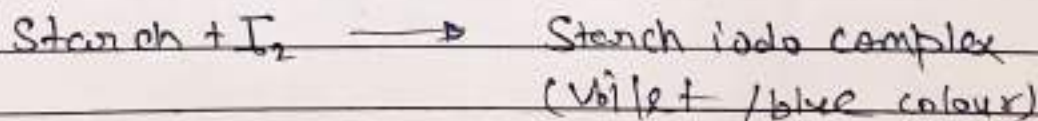
Ans When potassium iodide is added at pH (4 to 4.7) to copper sulphate solution, an equivalent amount of iodine is liberated & solution becomes brown.



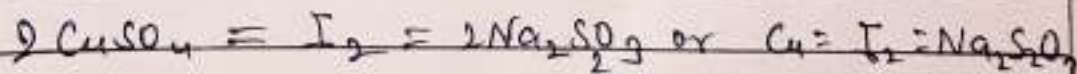
This liberated iodine is titrated with standard solution of sodium thiosulphate by using starch as an indicator



The liberated iodine absorbs on starch & gives deep blue or violet colour.



from the above reaction it follows that



Q3 Write chemical name & formula of Hypo Solution.
Ans Hypo solution is nothing but Sodium thiosulphate its chemical formula is $\text{Na}_2\text{S}_2\text{O}_3$
In aqueous form = $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$

Q4 How will you prepare $\text{N}_2\text{S}_2\text{O}_8$.
Explain the principle Q^3 .

Q5 Why pH should be maintained between 3 to 4 in this titration?

Ans Iodine remains stable & unreacted with hydroxide or oxygen.

Accurate stoichiometry in the redox reaction.
Reliable endpoint detection using starch.